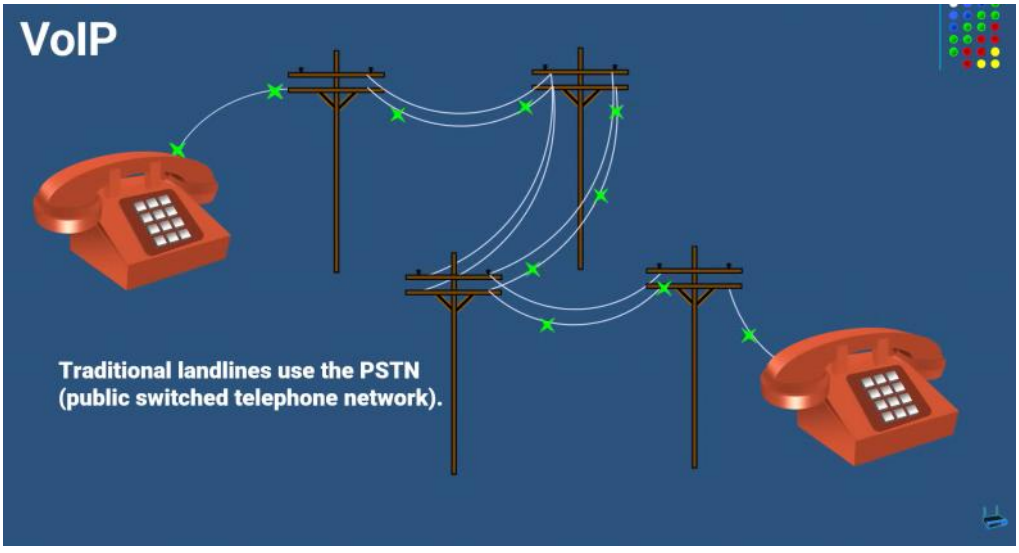
VoIP

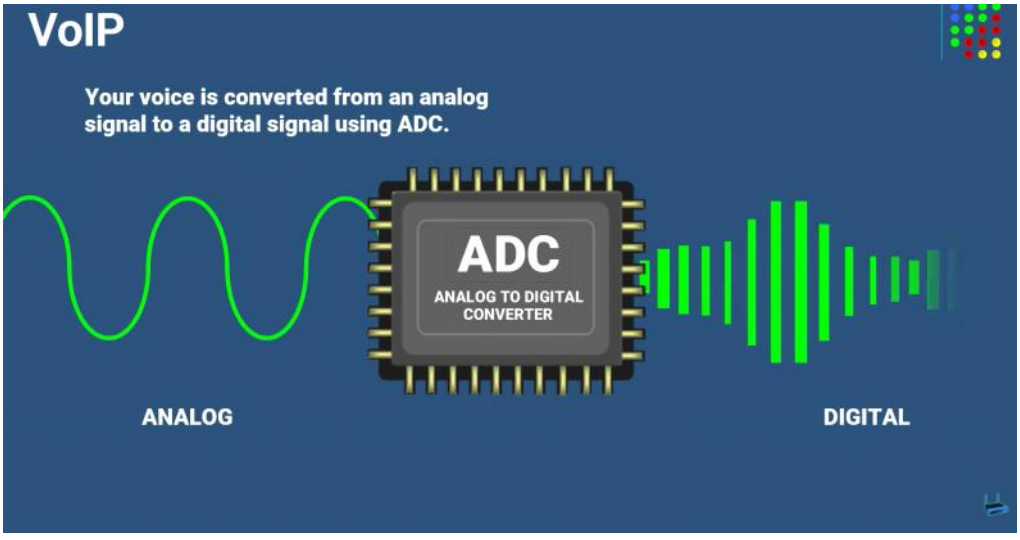
Tuesday, April 28, 2026 8:32 AM

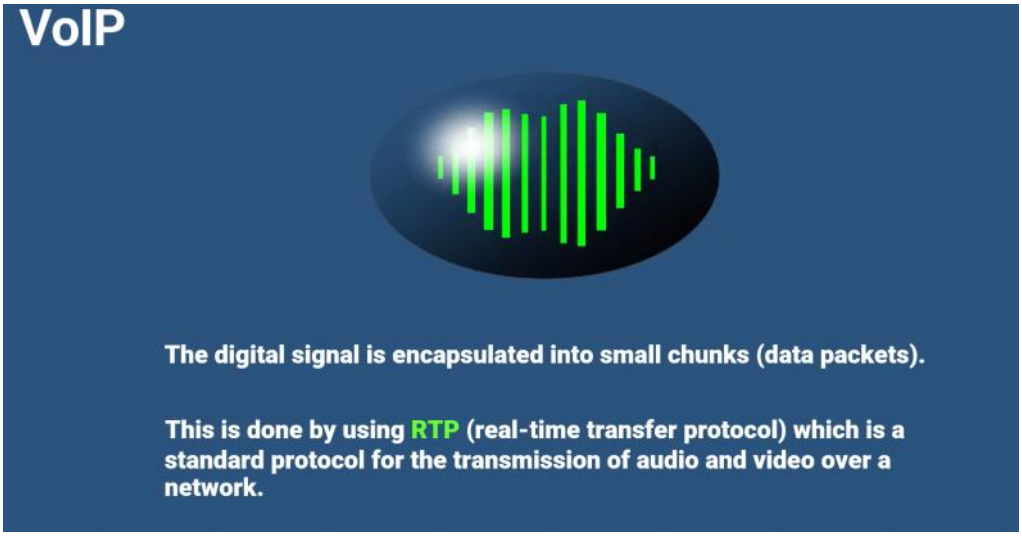
VoIP (Voice over Internet Protocol) is a technology that allows you to make phone calls over an internet connection instead of using traditional landline telephone systems.

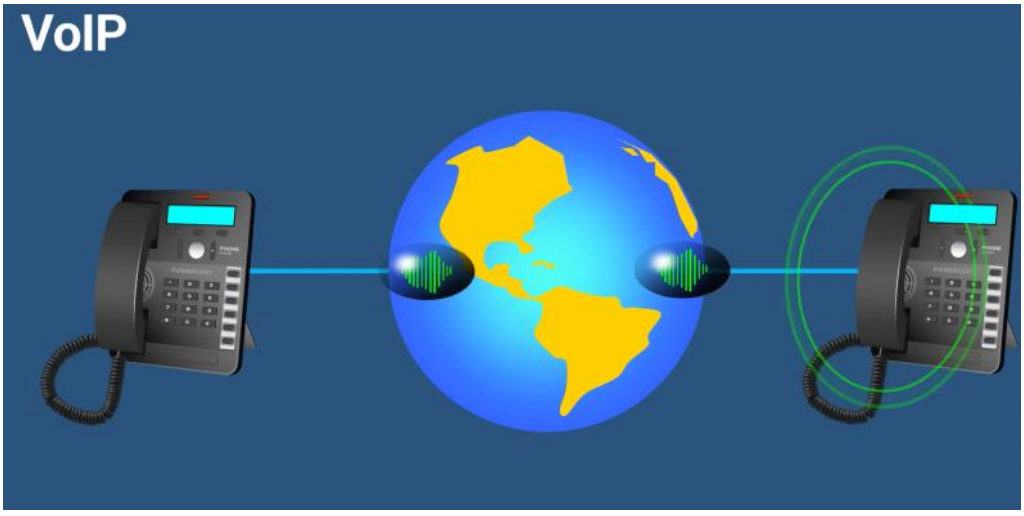


How it works:

- Initialization: A protocol called *SIP* (Session Initiation Protocol) maps phone numbers to IP addresses to connect calls.
- Conversion: An *ADC* (Analog to Digital Converter) changes your voice into digital signals.
- Transmission: These signals are broken into data packets and sent across the internet using *RTP* (Realtime Transfer Protocol), where they are reassembled for the recipient.







Equipment options:

- VoIP Phone: A specialized device that plugs directly into a router via an Ethernet port.
- Analog Telephone Adapter: Allows you to use a traditional landline phone for internet-based calls.
- Softphone: Software applications installed on computers, tablets, or smartphones that turn those devices into phones.

Benefits:

- Cost-effective: Generally cheaper for long-distance and international calls, and eliminates the need for expensive on-site *PBX* hardware.
- Flexibility: Enables employees to make calls from almost anywhere with an internet connection.
- Features: Includes advanced tools like conference calling, call recording, and voicemail management.

Disadvantages:

- Reliability: Requires a stable internet connection; poor quality can occur if the connection is unstable.
- Power dependence: Services typically do not work during power outages.
- Compatibility: Some security and alarm systems may require specific testing for compatibility with certain providers.